

# Sayantana Bhattacharya

POSTDOCTORAL RESEARCHER · DATA SCIENTIST

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## Interest

Computational Social Science, Collective Action, Machine Learning, Natural Language Processing

## Current Position

### Institute of Data Democracy & Politics, The George Washington University

Washington DC, USA

POSTDOCTORAL RESEARCHER

May. 2026 - Present

- Researching grievance measurement in social media by developing a Grievance Measurement Index (GMI) to quantify expressions of injustice, blame, collective identity, emotional intensity, and mobilization intent in online posts.
- Designing a post-level and movement-level scoring framework to analyze how grievance narratives evolve over time and whether sudden grievance surges can predict collective action or protest events.
- Combining social movement theory, framing theory, and computational text analysis to study how online grievance expression relates to political mobilization.

## Academic Experience

### Collaboratorium for Social Media and Online Behavioral Studies

Little Rock, AR

GRADUATE RESEARCH ASSISTANT / PHD CANDIDATE

Aug. 2022 - December 2025

- **Project:** “Developing Socio-computational Approaches to Mitigate Collective Action-based Socio-cognitive Security Threats in a Multi-platform Multimedia-rich Information Environment”. The project is funded by the U.S. Army Research Office (ARO) Social and Cognitive Networks Program. Award number: W911NF-24-1-0078
- As a part of Collective Action Research study, currently working on analyzing Social Movement and Election related discourse through the lens of Socio-Technical/AI Framework.
- Spearheaded research efforts within the Indo-Pacific initiative focusing on Malaysian political discourse, resulting in actionable insights into voter behavior patterns during 2022 election.
- Amid Taiwan's 2024 presidential election, research into the anti-disinformation campaign leverages large language models to decode role symbols that articulate political identities and strategic narratives.

## Education

### University of Arkansas at Little Rock

Little Rock, AR, USA

PHD IN COMPUTER & INFORMATION SCIENCE

AUG. 2022 – MAY 2026

### University of Arkansas at Little Rock

Little Rock, AR, USA

MASTER OF SCIENCE IN INFORMATION SCIENCE

AUG. 2022 – MAY 2024

### University Institute of Technology, The University of Burdwan

Bardhaman, WB, India

BACHELOR OF ENGINEERING IN INFORMATION TECHNOLOGY

2011 – 2015

## Experience

### Data Science Intern

Little Rock, AR, USA

ARKANSAS DEPT. OF EDUCATION

May. 2023 - Aug. 2023

- Contributed to the deployment of “Arkansas LEARN,” ensuring smooth integration and post-launch stability.
- Served as Backend and Frontend developer for different module in LEARN application.
- Conducted detailed analysis and addressed critical bugs, enhancing system reliability.

### Application Developer Senior Analyst

Kolkata, WB, India

ACCENTURE

Aug. 2021 - Aug. 2022

- Developed end-to-end solution with Master Data Management (MDM) and Product User Interfaces.
- Participated in product-related bug fixing and coordinated with support teams by raising tickets in provided platforms.
- Delivered multiple deployments, meeting project deadlines and ensuring robust functionality.

## Associate Developer

COGNIZANT TECHNOLOGY SOLUTIONS

Kolkata, WB, India

May. 2016 - Aug. 2021

- Understanding domain (Healthcare, Banking/Finance and Insurance) methodologies and functionalities.
- Involved in large scale data analysis and data loading activities.
- Contributed to end-to-end solution with Master Data Management (MDM) and Product User Interfaces.
- Fixed identified issues and worked with applications to implement necessary code changes.
- Developed, modified, and reviewed testing components based on defined specifications and tests.
- Designed and implemented match configurations that improved data accuracy, enhancing the overall quality of data integration processes within Master Data Management (MDM) systems for more reliable outputs.

## Awards & Grants

- |      |                                                                                     |                    |
|------|-------------------------------------------------------------------------------------|--------------------|
| 2024 | <b>DCSTEM Travel Award</b> , Received on Fall 2024                                  | Little Rock, USA   |
| 2024 | <b>Graduate School Travel Award</b> , Received in Fall 2024                         | Little Rock, U.S.A |
| 2024 | <b>Student Research and Creative Works Expo</b> , Received 3rd Prize on Spring 2024 | Little Rock, U.S.A |
| 2023 | <b>Student Research and Creative Works Expo</b> , Received 1st Prize on Spring 2023 | Little Rock, U.S.A |

## Presentation

### 2024/2025 Americas Conference on Information Systems (AMCIS)

Saltlake City, Utah

PRESENTED PAPER ON COLLECTIVE ACTION FRAMEWORK

Aug. 2024/2025

- **Paper:** A Computational Approach to Analyze Identity Formation: A Case Study of Brazil Insurrection

### 17th International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation.)

Pittsburg, Pennsylvania

PRESENTED PAPER ON INFLUENTIAL AGENTS IN SOCIAL NETWORK

Aug. 2024

- **Paper:** Ablation studies in protest networks: The role of influential agents in shaping protests

### Hawaii International Conference on System Sciences(HICCS)

Big Island, Hawaii

PRESENTED PAPER ON ROLE OF KEY USERS IN CONTRIBUTING TO COLLECTIVE IDENTITY FORMATION

Jan. 2025

- **Paper:** The Amplifiers of Dissent: Examining Influence of Key Users and Content Modality on Collective Actions

### Advances in Social Networks Analysis and Mining-ASONAM-2025

Niagara Falls, Canada

PRESENTED PAPER ON PERSUASIVE POWER OF VISUAL ELEMENTS IN STRATEGIC COMMUNICATION

August. 2025

- **Paper:** The Persuasive Power of Visual Elements in Strategic Communication

## Paper Review & PC Member

SBP BRIMS 2024, AMCIS 2024, HICCS 2024, COMPLEX NETWORK 2024, AIES 2025, JOURNAL OF SOCIAL NETWORK ANALYSIS AND MINING, COMPUTERS IN HUMAN BEHAVIOR, SOCIAL SCIENCE AND HUMANITIES, FRONTIERS IN POLITICAL SCIENCE - POLITICAL PARTICIPATION

## Skills

**Programming Languages**, Python, Javascript

**Hard Skills**, Machine Learning, Data Visualization, Statistical Analysis

**Soft Skills**, Problem Solving, Critical Thinking, Interpersonal Skills

- **S Bhattacharya**, B Spann, and N Agarwal. Solidarity to Storming: Assessing the Socio-Technical Factors Behind Modern Social Movements. European Conference of Information System (ECIS), 2024. Paphos, Cypress
- **S Bhattacharya**, N Agarwal, and D Poudel. Ablation studies in protest networks: The role of influential agents in shaping protests. International Conference on Social Computing, Behavioral-Cultural Modeling, SBP-BRIMS 2024. Pittsburg, Pennsylvania
- **S Bhattacharya**, B Spann, and N Agarwal. A Computational Approach to Analyze Identity Formation: A Case Study of Brazil Insurrection. American Conference of Information System (AMCIS), 2024. Saltlake City, Utah
- **S Bhattacharya**, N Agarwal, and D Poudel. The Amplifiers of Dissent: Examining Influence of Key Users and Content Modality on Collective Actions. 58th Hawaii International Conference on System Sciences (HICCS-58). Big Island, Hawaii.
- **S Bhattacharya**, N Agarwal, and D Poudel. Analyzing the Impact of Symbols in Taiwan's Anti-Disinformation Campaign on TikTok during Elections. Journal of Social Network Analysis and Mining (SNAM) 14.
- **S Bhattacharya**, N Agarwal. Decoding Digital Dissent through Focal Network Structures. 13th International Conference on Complex Networks & Their Applications. Istanbul, Turkey
- D Poudel, **S Bhattacharya**, N Agarwal. Digital Symbols and Democratic Trust. Americas Conference in Information System. Montreal, Canada - **Top 25% paper**
- **S Bhattacharya**, N Agarwal, and D Poudel. The Viral Nature of Symbols: Epidemiological Modeling of Visual Elements in Digital Information Campaigns. CySoc 2025 (6th International Workshop on Cyber Social Threats at ICWSM 2025)
- **S Bhattacharya**, N Agarwal, and D Poudel. The Persuasive Power of Visual Elements in Strategic Communication. The 17th International Conference on Advances in Social Networks Analysis and Mining-ASONAM-2025
- **S Bhattacharya**, N Agarwal, and D Poudel. The Persuasive Power of Visual Elements in Strategic Communication. International Conference on Social Computing, Behavioral-Cultural Modeling. Pittsburg, Pennsylvania-SBP-BRIMS 2025 - **Best Paper Award**
- **S Bhattacharya** and N Agarwal. Analyzing Key Network Structures of 2022 Malaysian General Elections from the Lens of Instagram. The 14th International Conference on Social Media Technologies, Communication, and Informatics, 2024